

Web 3.0

Building the Semantic Web

You might have come across the term 'Semantic Web Applications' often, during talks about the future of Web apps. Check out what this is all about.

There are two aspects to the possible future of the Web; one is that it is definitely going to be more social and more people-centric, but the other important characteristic is that it's going to be more semantic in nature. The Web technologies that we have available today focus more on how pages are displayed and what content is displayed, rather than on the content itself. The semantic Web is a facet of technology that will allow this content to become meaningful for machines, and will enable them to process this content and help us share, combine and analyse content effectively. To make it clear, let me give you the example of Wikipedia. It has all the information in the world about everything related to rivers, but can I query it to find out about the longest river in India? I guess not, because I will have to search for it manually before retrieving that information, and there is no way to get this data programmatically from Wikipedia in its present form. Enter DBpedia, and you will be surprised to find that you can do much more than that. During the course of this article, we'll look at how this happens.

Why the Semantic Web?

But why go through all the hassle, you might ask? Well, natural language processing (NLP) is one field that could benefit from this. I'm not much of a fan of the Apple iPhone ecosystem, but we must agree that Siri is really an impressive feature. Having a machine that can easily interact with humans and answer random queries will be the next leap in the evolution of computing. Another example where semantic information might come in handy is in search engines, which can then more effectively derive what the relationships between Web pages are, and understand their context, thus providing better search results that are relevant to the search query entered by the user. Google allows for retrieving such semantic information, as you can see in Figures 1 and 2.

The Semantic Web stack

The Semantic Web stack is something that is more of a

theoretical topic, and you'll find fat books on the topic. But when it comes to actually implementing it, it's a little more challenging than it sounds—and to put it into practice is a little overwhelming. Of course, we will look at the challenges that it can face, during the course of this article.

The various technologies that come under the umbrella of the 'Semantic Web' have been under development, independently, for quite some time. Only after their recent standardisation have people realised that they are all part of the larger picture. Of these, the major technologies include:

- RDF (Resource Description Framework)
- RDFS (RDF Schema)
- OWL (Web Ontology Language)
- SPARQL (SPARQL Protocol and RDF Query Language)

These technologies collectively allow us to add semantic information to the vast number of existing Web pages. The aim recently has been to build upon the foundation of the Web as we know it, rather than to replace it altogether, because the former option is comparatively easier, when you consider the vast existing Web content that would be difficult to replace.

Resource Description Framework

RDF is a specification that deals with modelling or representation of data. Each set of information is represented in the form of subject-predicate-object syntax. There are a variety of formats in which this is serialised. Some of the popular ones are:

- RDF/XML
- RDFa
- N3
- Turtle

Again, these formats are only a subset of the many other notations available, which are under development. An example of how we would represent semantic data in the form of RDF/XML, which is the most common format, is as follows:

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<?xml version="1.0"?><br>
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